

A BibTeX entry for LaTeX users is

```
@Manual{,
  title = {R: A Language and Environment for Statistical
Computing},
  author = {{R Core Team}},
  organization = {R Foundation for Statistical Computing},
  address = {Vienna, Austria},
  year = {2021},
  url = {https://www.R-project.org/},
}
```

We have invested a lot of time and effort in creating R, please cite it when using it for data analysis. See also 'citation("pkgname")' for citing R packages.

History

R is an alternate implementation of the S programming language. S is a statistical programming language created by John Chambers in 1976 at Bell Laboratories (previously AT&T). Rick Becker and Allan Wilks of Bell Laboratories have also worked on the initial releases of S. The S programming language is dynamically and strongly typed, and supports both the imperative and object-oriented styles

of programming. Most of the S code actually runs without alterations on R.

In 1991, Ross Ihaka and Robert Gentleman at the University of Auckland, New Zealand, wrote an alternative implementation to the S programming language, which was promoted as the R programming language in 1993. The R project was officially released in 1995 as Free/Libre and Open Source Software (FOSS) and is now maintained by the R core team.

The 'R Foundation for Statistical Computing' or the 'R Foundation' was created by the R core team to facilitate the development of the R programming language, and its tools and ecosystem. It also offers support for all users, developers and organisations using R in the community and for commercial purposes. It is responsible for the copyright of the R software and documentation. The foundation also conducts meetings and conferences regularly, and its annual conference is called useR!.

In the next article in this series, we will go over the syntax and semantics of the R programming language. **END** 

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can be listed. For the example code explained above, the file is located in `sklearn_elasticnet_wine/MLproject`.

```
name: tutorial
conda_env: conda.yaml
entry_points:
  main:
    parameters:
      alpha: {type: float, default: 0.5}
      l1_ratio: {type: float, default:
0.1}

    command: "python train.py {alpha}
{l1_ratio}"
```

`conda.yaml` lists the dependencies:

```
name: tutorial
channels:
  - defaults
dependencies:
  - python=3.6
```

```
- pip
- pip:
  - scikit-learn==0.23.2
  - mlflow>=1.0
```

This project can be executed with specific parameter values, as shown below:

```
mlflow run sklearn_elasticnet_wine -P
alpha=0.48
```

Serving the model

It can be observed from the sample code that it saves the model as an artifact with the command code:

```
mlflow.sklearn.log_model(lr, "model")
```

The artifacts can be explored through the UI by clicking the specific date. A sample is shown in Figure 7.

The model can be served with `mlflow models serve`.

MLflow makes the ML application development life cycle efficient. Though there are alternatives available, a salient feature of MLflow is its simplicity. The universal support for ML libraries makes it an important platform. **END** 

References

- [1] <https://mlflow.org/>
- [2] <https://mlflow.org/docs/latest/tutorials-and-examples/tutorial.html>
- [3] <https://www.youtube.com/watch?v=x3cxvsUFVZA&t=1020s>

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